Q.1. Prove that $\sup A = 3$ for $A = \{x \in \mathbb{R} : x < 3\}$, using ε-characterization of supremum.

Soln :- To prove that

$\sup A = 3$, for the set $A = \{x \in \mathbb{R} : x < 3\}$,

* 3 is an upper bound of A :-

By definition of the set A

$$x < 3 \; ; \; \forall x \in A$$

Hence, 3 is an upper bound of A

3 is an least upper bound of A :-

For an $\varepsilon > 0$, $\exists\, x_i \in A$ such that,

$$3 - \varepsilon < x$$

Consider the element

$$x_0 = 3 - 3/2 \in A$$

Hence, $3 - \varepsilon < x_0$

$\therefore$ 3 is an least upper bound of A

$\therefore \sup A = 3$

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Q2. Using the Archimedean Property, Prove that $\inf\{\frac{1}{n} : n \in \mathbb{N}\} = 0$

Soln let $A = \{\frac{1}{n} : n \in \mathbb{N}\}$,

To prove that $\inf A = 0$

0 is an lower bound of A :-

Since, $n > 0$

so, $\frac{1}{n} > 0$, $\forall\ n \in \mathbb{N}$

i.e $n > 0$, $\forall\ n \in A$

∴ 0 is a lower bound of A

0 is an Greatest lower bound of A :

let $\varepsilon > 0$ be any $\mathbb{R}$

So, by Archimedean Property, we have

$\exists\ n \in \mathbb{N}$ s.t

$\frac{1}{n} < \varepsilon$ i.e $0 + \varepsilon > n$, $\exists\ n \in A$

So, 0 is an Greatest lower bound of A

∴ $\inf\{\frac{1}{n} : n \in \mathbb{N}\} = 0$

Q.3 Prove the density of Q in $\mathbb{R}$ for any $a < b$ in $\mathbb{R}$

There exist $r_0 \in Q$ with $a < r_0 < b$.

Solution :- without loss of Generality, let $a > 0$.

Since, $b-a > 0$, we get $\exists\, n \in \mathbb{N}$ by Anchimedean Propasty

i.e $\frac{1}{n} < b-a$

i.e $1 < bn - an$

i.e $1 + an < bn \longrightarrow (1)$

Since $a > 0$, we have $na > 0$

Let $\exists m \in \mathbb{N}$ then we have,

$$m-1 \leq na < m$$

i.e $m \leq na+1 < m+1 \longrightarrow (2)$

From (1) and (2) we have,

$$m < na+1 < bn \longrightarrow (3)$$

Also we have, $na < m \longrightarrow (4)$

From (3) and (4) we have

$$na < m < bn$$

Thus, the, rational number to $:= \frac{m}{n}$

satisfies, $a < \frac{m}{n} < b$

(Q.4) Let A, B be nonempty subsets of $\mathbb{R}$, bounded below with $A \subseteq B$. Prove that $\sup A \leq \sup B$

Solution:- By definition, $\sup B$ is an upper bound of B

i.e $x \leq \sup B, \forall x \in B$

Since $A \subseteq B$, Then

$x \leq \sup B, \forall x \in A$

Hence, $\sup B$ is an upper bound of A

Because $\sup A$ is least upper bound of A

i.e $\sup A \leq \sup B$

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(Q.5) For nonempty bounded sets, $A, B \subseteq \mathbb{R}$, let $A+B = \{a+b : a \in A, b \in B\}$. Show that ~~$\sup A \leq \sup B$~~. $\sup(A+B) = \sup A + \sup B$

Solution Let $\alpha = \sup A$ and $\beta = \sup B$, Since A and B are non-empty and bounded above, $\alpha, \beta \in \mathbb{R}$ exist by Completeness property.

and To show that $\sup(A+B) = \alpha + \beta$

Step-1:- $\alpha + \beta$ is an upper bound of $A+B$

By definition of supremum;

$\alpha = \sup A$ i.e $a \leq \alpha, \forall a \in A$

and, $B = \sup B$ i.e $b \leq \beta, \forall b \in B$

Adding these inequalities, we get,

$$a+b \leq \alpha+\beta \quad , \forall\ a \in A\ ,\ b \in B$$

~~Since~~, Since every element in $A+B$ is of the form $a+b$

$\therefore \alpha+\beta$ is an upper bound of $A+B$

Step-II : $\alpha+\beta$ is an least upper bound of $A+B$

let $\varepsilon > 0$,

since, $\alpha = \sup A$, $\alpha - \frac{\varepsilon}{2}$ is not an upper bound of A

$\therefore\ \exists\ a_0 \in A$ such that,

$$\alpha - \varepsilon/2 < a_0$$

Similarly, since, $\beta = \sup B$, $\exists\ b_0 \in B$ such that

$$\beta - \varepsilon/2 < b_0$$

Adding these two inequalities, we get,

$$(\alpha - \varepsilon/2) + (\beta - \varepsilon/2) = \alpha+b-\varepsilon < a_0+b_0\ ,\ \exists\ a_0 \in A\ ,\ b_0 \in B$$

i.e $\alpha+\beta-\varepsilon < a_0+b_0$, $\exists\ a_0+b_0 \in A+B$

Hence $\alpha+\beta$ is an least upper bound of $A+B$

$\therefore\ \sup(A+B) = \alpha+\beta$

i.e $\sup(A+B) = \sup A + \sup B$

Q6 Prove that between any two real numbers there exists an irrational numbers.

Solution let, if x and y are any two real numbers with $x < y$ then we want to prove that, there exist an irrational number z such that $x < z < y$.

Apply the Density theorem to the Real numbers $x/\sqrt{2}$ and $y/\sqrt{2}$ we obtain, a ration number $r_0 \neq 0$, such that,

$$\frac{x_0}{\sqrt{2}} < r_0 < \frac{y}{\sqrt{2}}$$

Then $z := r_0\sqrt{2}$ is irrational and satisfies

$$x < z < y$$

(Q.7) Determine sup and inf of $E := \{x \in \mathbb{R} : x^2 < 2\}$ and explain how this connect to the existence of $\sqrt{2}$ as a real number.

Solution :- determine sup(E) and inf(E).

Given that, $E := \{x \in \mathbb{R} : x^2 < 2\}$

i.e $x^2 - 2 < 0$

i.e $(x - \sqrt{2})(x + \sqrt{2}) < 0$

i.e $-\sqrt{2} < x < \sqrt{2}$, $\forall\ x \in E$

Hence, $\sup(E) = \sqrt{2}$

$\inf(E) = -\sqrt{2}$

Connection to the Existence of $\sqrt{2} \in \mathbb{R}$;;

The existence of $\sqrt{2}$ in $\mathbb{R}$ relies directly on the Completeness Property of the real numbers.

→ Boundedness in $\mathbb{R}$: The set E is non-empty (since $1 \in E$) and bounded above,

→ Guarantee of Supremum ; By the Completeness property every non empty subset of $\mathbb{R}$ bounded above must have a supremum in $\mathbb{R}$. Therefore, $\alpha = \sup(E)$ exist as a Real numbers.

→ Proving $\alpha^2 = 2$

- if $\alpha^2 < 2$, for any $\varepsilon > 0$ such that $(\alpha+\varepsilon)^2 < 2$, which means $\alpha+\varepsilon \in E$, contradicting that α is an upper bound.
- if $\alpha^2 > 2$, for any $\varepsilon > 0$ such that $(\alpha-\varepsilon)^2 > 2$, implying $\alpha-\varepsilon$ is still an upper bound of E. contradicting that α is the least upper bound.
- By Trichotomy, we must have $\alpha^2 = 2$

Thus defining $\sqrt{2}$ as $\sup(E)$ provides a rigorous, constructive proof that a unique positive real number whose square equals 2 exist in $\mathbb{R}$ - a property that fails in $\mathbb{Q}$, where the set $\{x \in \mathbb{Q} ; x^2 < 2\}$ has no supremum.

(9.8) Let A be a nonempty subset of $\mathbb{R}$ bounded below, and Let $-A = \{-a : a \in A\}$. Prove that $\inf A = -\sup(-A)$.

Solution:

Given that

$$-A = \{-a : a \in A\}$$

To prove that $\inf A = -\sup(-A)$

Let $m = \inf A$

$$m \leq a \quad , \forall\, a \in A$$

multiplying both side by -1, then,

$$-m \geq -a \quad , \forall\, a \in A$$

i.e $-a \leq -m \quad , \forall\, -a \in -A$

$\therefore$ $-m$ is an upper bound of $-A$

For any $\varepsilon > 0$, since $m = \inf A$

so, $m + \varepsilon > a \quad , \exists\, a \in A$

multiplying by -1 then, we have,

$$-m - \varepsilon < -a \quad , \exists\, a \in A$$

i.e $(-m) - \varepsilon < -a \quad , \exists\, -a \in -A$

$-m$ is an least upper bound of $-A$

$\therefore$ $\sup(-A) = -m$

i.e $\sup(-A) = -\inf A \Rightarrow \inf A = -\sup(-A)$ #